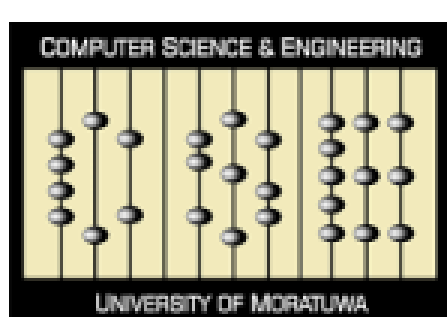




Evaluating LLMs' Contextual Understanding with Knowledge Graphs

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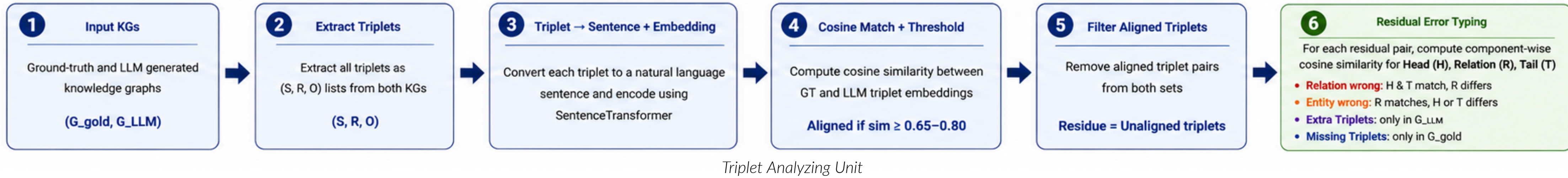
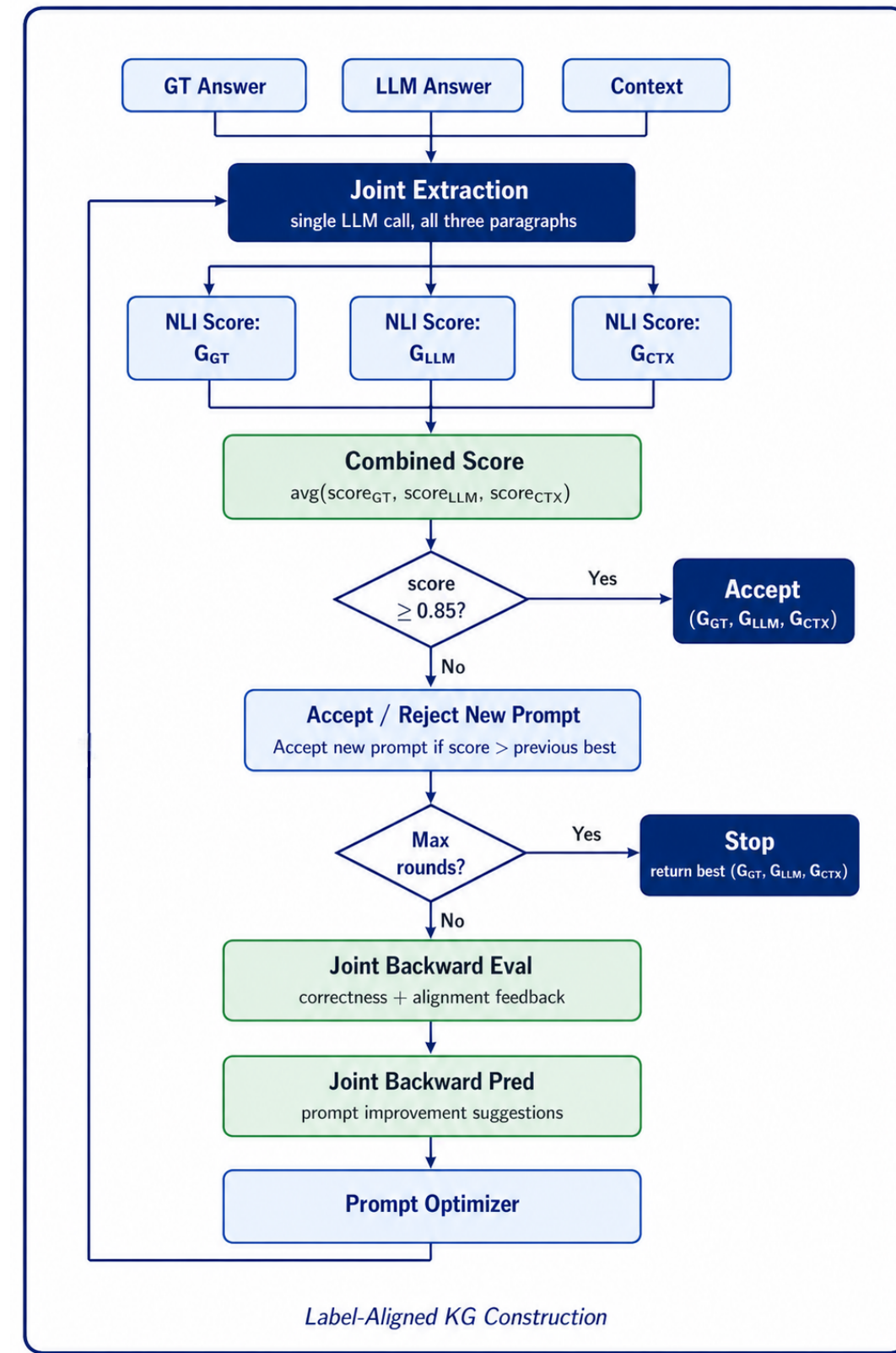
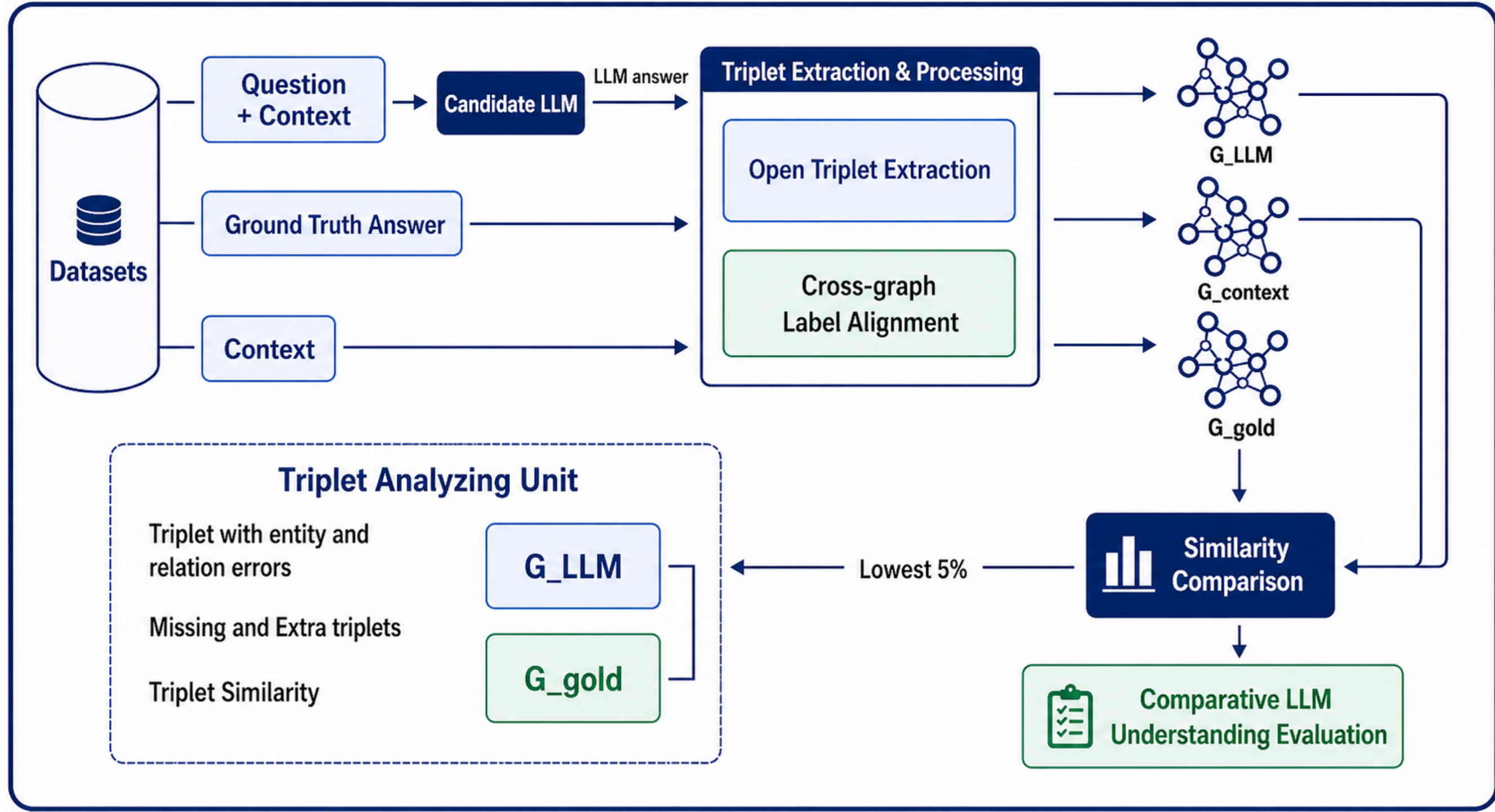
Problem Statement

LLMs answer questions well, but their **depth of contextual understanding** remains unclear. Existing evaluations focus on **surface correctness**, missing which context aids or misleads, where reasoning fails, and how much is truly understood. In QA, where grounding is essential, we need **systematic, interpretable** methods to measure and visualise understanding beyond surface metrics.

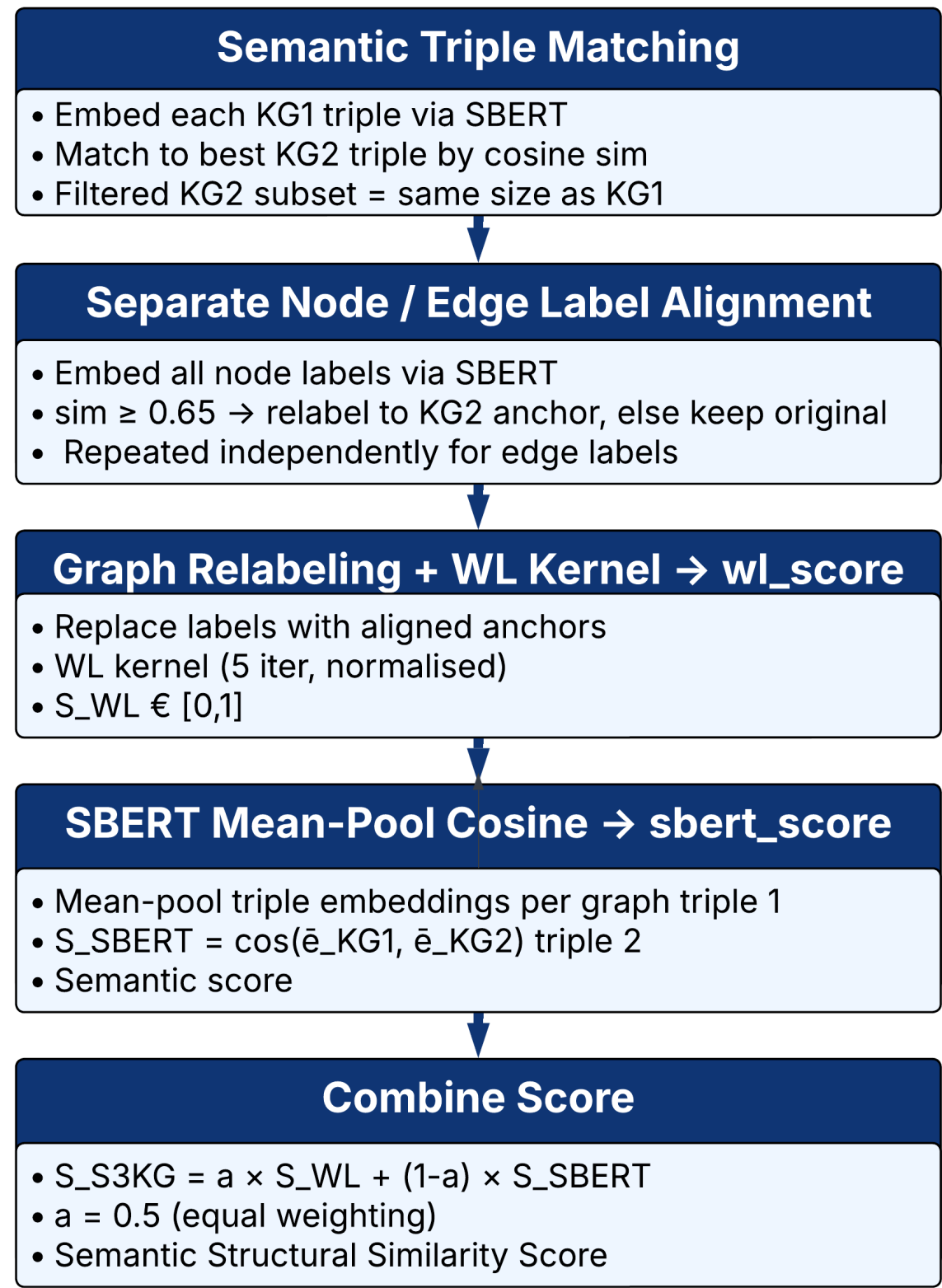
Research Objectives

- Develop & validate a KG-based methodology to evaluate LLM contextual understanding via entity/relation extraction and graph similarity scoring.
- Detect fact deviations (missing, substituted, spurious entities/relations) and
- Establish a reproducible framework adaptable to other datasets and tasks.

Methodology



S3KG: Semantic Structural Similarity



LLM Contextual Understanding Evaluation

Three KGs per QA instance compared via S3KG ($\alpha=0.5$):

- GoldSim — factual accuracy
- CtxSim — contextual faithfulness
- CUS — harmonic mean of GoldSim & CtxSim:

$$CUS(i) = \frac{2 \cdot GoldSim(i) \cdot CtxSim(i)}{GoldSim(i) + CtxSim(i)}$$

Dataset	Model	GoldSim	CtxSim	CUS
MesaQA	Gemma-7B	0.6889	0.7031	0.6774
	Llama-2-7B	0.6570	0.7236	0.6752
	Mistral-7B	0.6491	0.7356	0.6780
	Falcon-7B	0.6036	0.6458	0.6063
PubMedQA	Gemma-7B	0.5235	0.6401	0.5587
	Llama-2-7B	0.5220	0.6567	0.5651
	Mistral-7B	0.5138	0.7331	0.5923
	Falcon-7B	0.4541	0.5560	0.4800

Mistral-7B leads CUS
Best on both datasets
(0.678 / 0.592)
CtxSim > GoldSim
All 8 model-dataset pairs
≈ 10 pp domain gap
MesaQA vs. PubMedQA
Falcon-7B weakest
Across all metrics

Benchmarking

	F1 Score									
	HBPC	PRMS	STN32	C400	Conts	Final	KG-Perturbed	Glots	Onp	Anti-C:
S3KG	0.69	0.77	0.79	0.93	0.83	0.77	0.89	0.81	0.87	
ROUGE-1	0.74	0.68	0.72	0.63	0.73	0.74	0.78	0.74	1	
ROUGE-2	0.72	0.71	0.68	0.62	0.71	0.72	0.78	0.75	0.68	
ROUGE-L	0.73	0.73	0.70	0.78	0.72	0.72	0.78	0.78	0.73	
BLEU	0.69	0.72	0.67	0.61	0.71	0.71	0.78	0.74	0.67	
BERTScore	0.76	0.69	0.68	0.82	0.76	0.74	0.82	0.74	0.75	
MuLMP	0.72	0.69	0.81	0.88	0.77	0.80	0.78	0.77	0.82	
SBERTS	0.77	0.67	0.85	0.88	0.77	0.85	0.73	0.80	0.76	

	AUROC									
	HBPC	PRMS	STN32	C400	Conts	Final	KG-Perturbed	Glots	Onp	Anti-C:
S3KG	0.67	0.80	0.83	0.97	0.88	0.80	0.94	0.89	0.89	
ROUGE-1	0.78	0.49	0.75	0.62	0.73	0.74	0.81	0.78	1	
ROUGE-2	0.72	0.72	0.66	0.69	0.75	0.71	0.83	0.79	0.77	
ROUGE-L	0.76	0.81	0.71	0.81	0.72	0.72	0.89	0.89	0.81	
BLEU	0.68	0.75	0.64	0.88	0.71	0.71	0.82	0.79	0.74	
BERTScore	0.82	0.70	0.64	0.92	0.79	0.76	0.87	0.80	0.85	
MuLMP	0.75	0.64	0.88	0.94	0.79	0.84	0.82	0.81	0.80	
SBERTS	0.82	0.67	0.93	0.94	0.81	0.90	0.75	0.87	0.81	

S3KG Benchmarking: Fluency $\neq$ factual faithfulness, these datasets are built to catch KG-level errors that Traditional Metrics are blind to.

Key findings:

- S3KG leads on 7/9 datasets in both F1 and AUROC
- Biggest gains on KG-Perturbed sets; structural reasoning matters most
- sent-T5 closest rival but collapses on relational datasets
- ROUGE-1 artefact on Wiki-Swap exposes surface overlap unreliability

KGC Benchmarking: Label alignment measured via Jaccard similarity across extracted graphs; extraction accuracy evaluated using strict F1 against reference triplets.

Method	Strict F1	Label Jaccard	H
KRPO (single-graph)	0.707	0.426	0.530
Ours	0.611	0.572	0.591
KEA Baseline	0.518	0.649	0.576

Triplet Unit Benchmarking: Gold triplets labeled by 3 medical-student annotators.

Metric	KEA	Ours	Agreement	Value
Micro F1	0.836	0.895	All-3 agree	0.9433
Macro F1	0.622	0.782	Pairwise F1	0.9708–0.9882

Conclusion

We present **CUS**, a reproducible metric for LLM contextual understanding as the combination of factual accuracy and the contextual faithfulness.

Our KGC method provides the most balanced performance across accuracy and label consistency for downstream multi-graph comparison tasks.

S3KG achieves top-ranked performance, outperforming existing methods on context-rich datasets.

Mistral-7B demonstrates the strongest contextual understanding among same-parameter models.

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